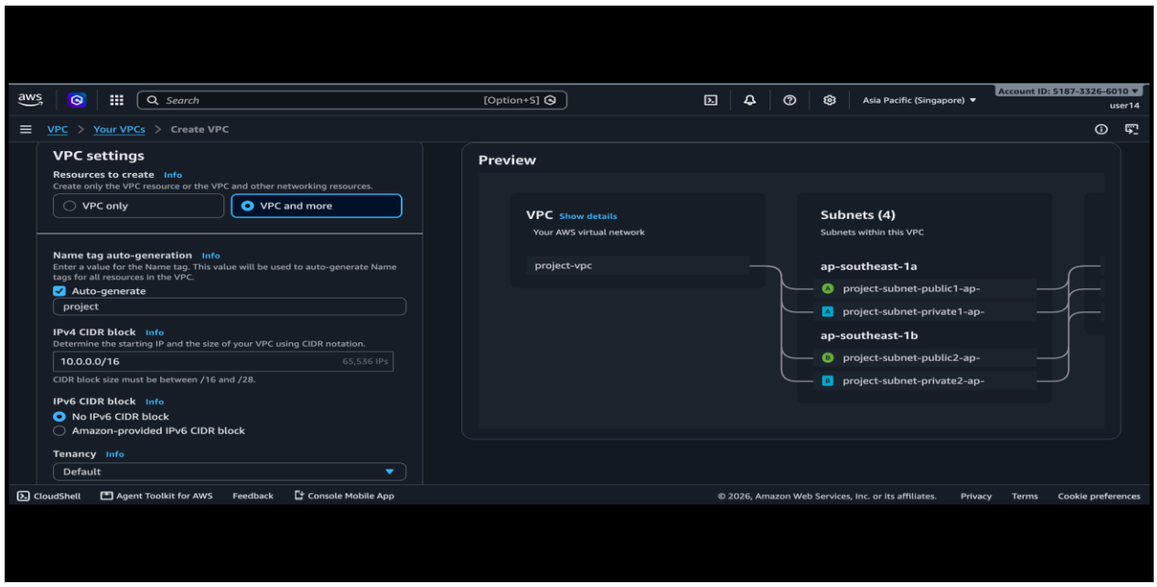
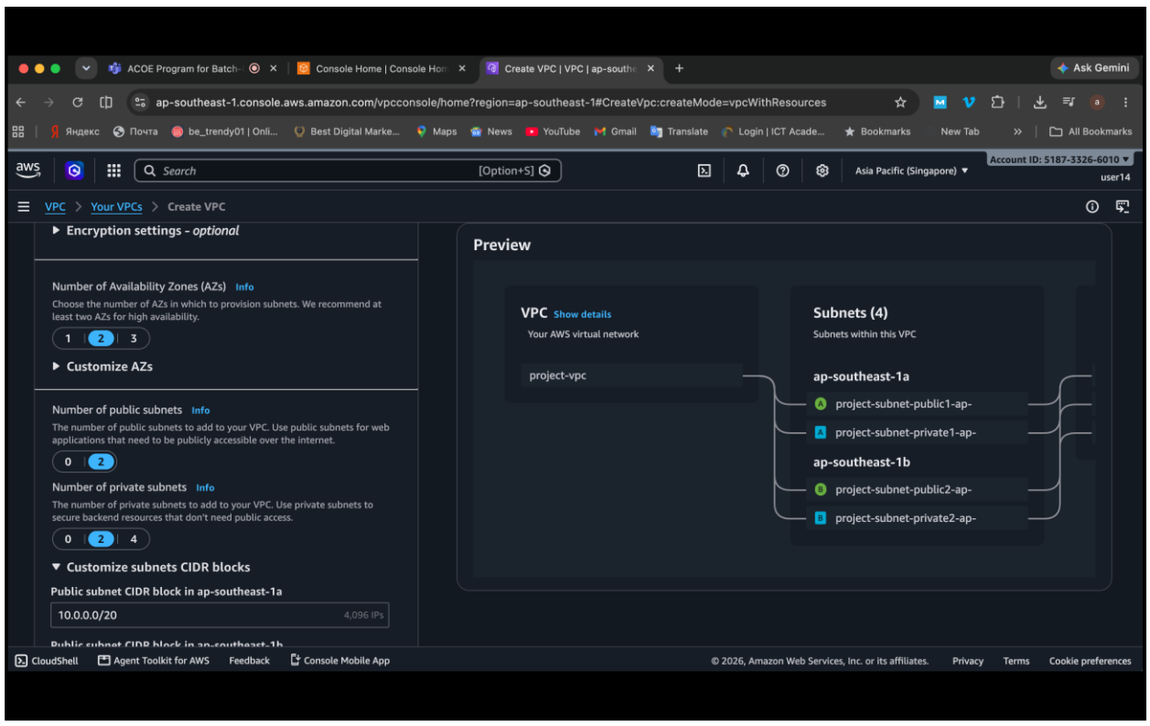
Case study -
Two tier, Highly Available and Highly Scalable architecture.

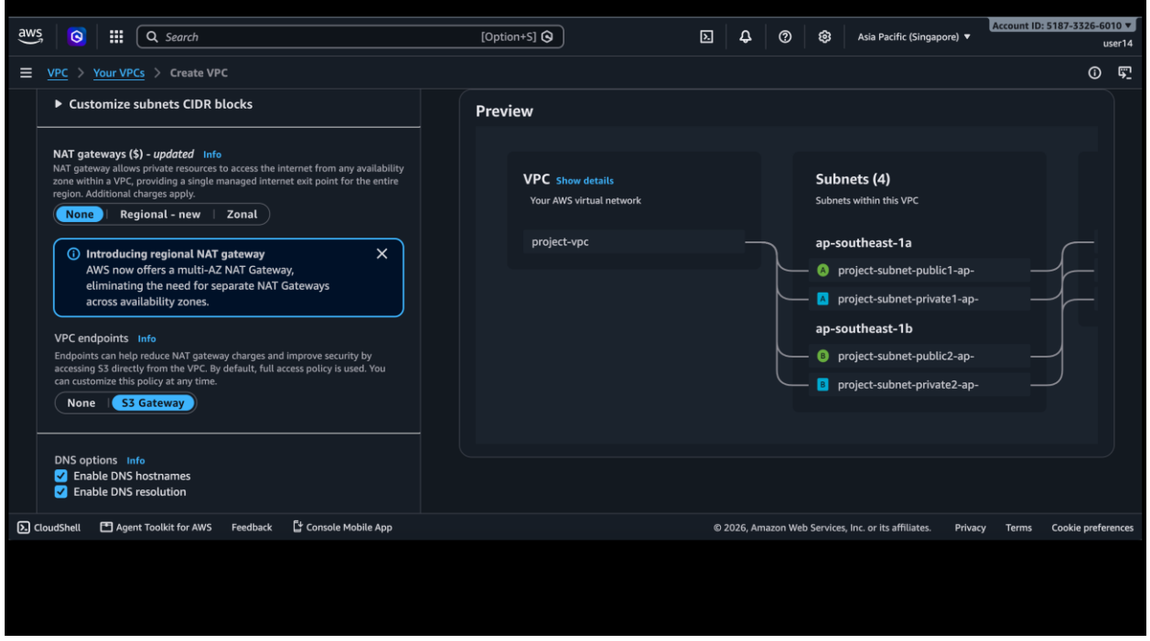
Problem statement solution

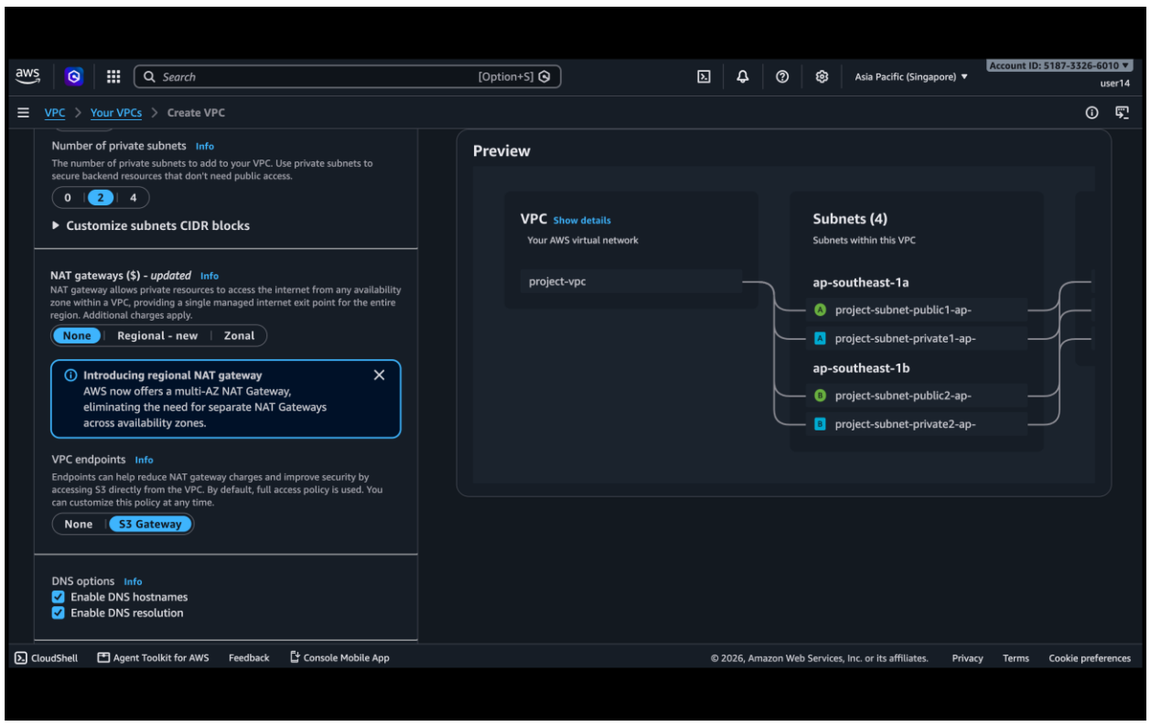
Task 1: Create a VPC

create a VPC, **CIDR blocks**, **2 Private Subnets**, **2 Public Subnets, Entries** in the Route table will get automatically updated.



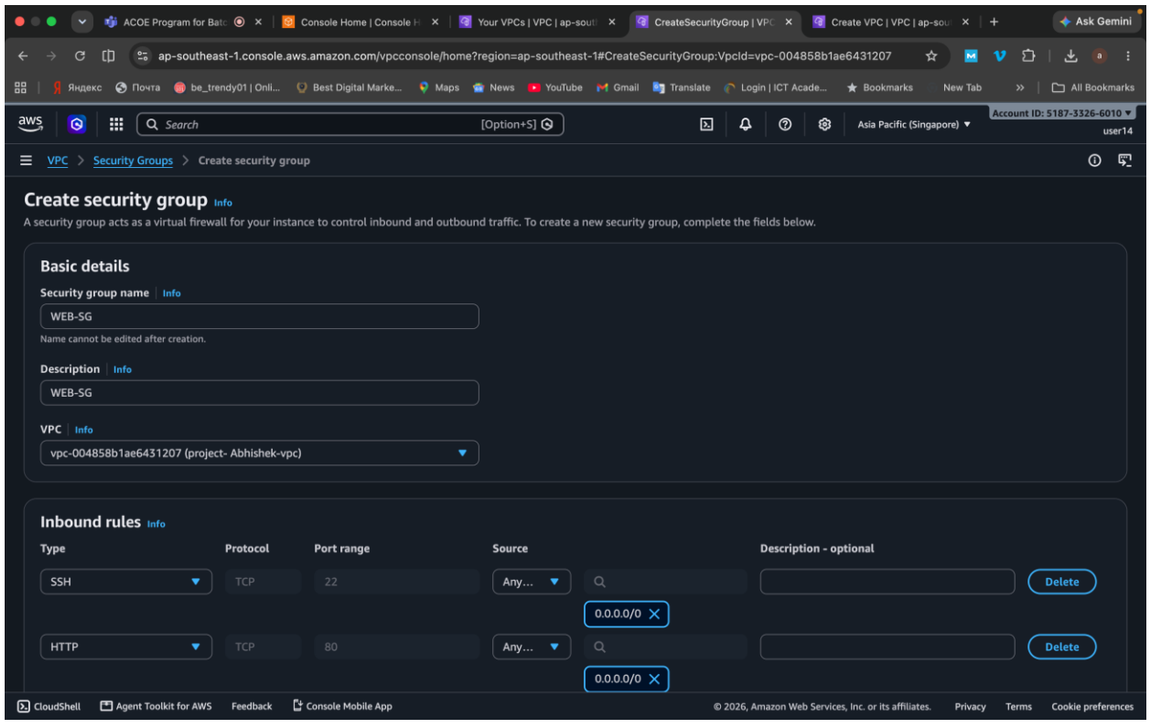

ACOE Program for Batch
Console Home | Console Hom
Create VPC | VPC | ap-southe
Ask Gemini
ap-southeast-1.console.aws.amazon.com/vpcconsole/home?region=ap-southeast-1#CreateVpc:createMode=vpcWithResources
Яндекс
Почта
be_trendy01 | Onli...
Best Digital Marke...
Maps
News
YouTube
Gmail
Translate
Login | ICT Acade...
Bookmarks
New Tab
All Bookmarks
Search
[Option+S]
Asia Pacific (Singapore)
Account ID: 5187-3326-6010
user14
VPC > Your VPCs > Create VPC
Encryption settings - optional
Number of Availability Zones (AZs) Info
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.
1 2 3
Customize AZs
Number of public subnets Info
The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.
0 2
Number of private subnets Info
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.
0 2 4
Customize subnets CIDR blocks
Public subnet CIDR block in ap-southeast-1a
10.0.0.0/20
4,096 IPs
Preview
VPC Show details
Your AWS virtual network
project-vpc
Subnets (4)
Subnets within this VPC
ap-southeast-1a
project-subnet-public1-ap-
project-subnet-private1-ap-
ap-southeast-1b
project-subnet-public2-ap-
project-subnet-private2-ap-
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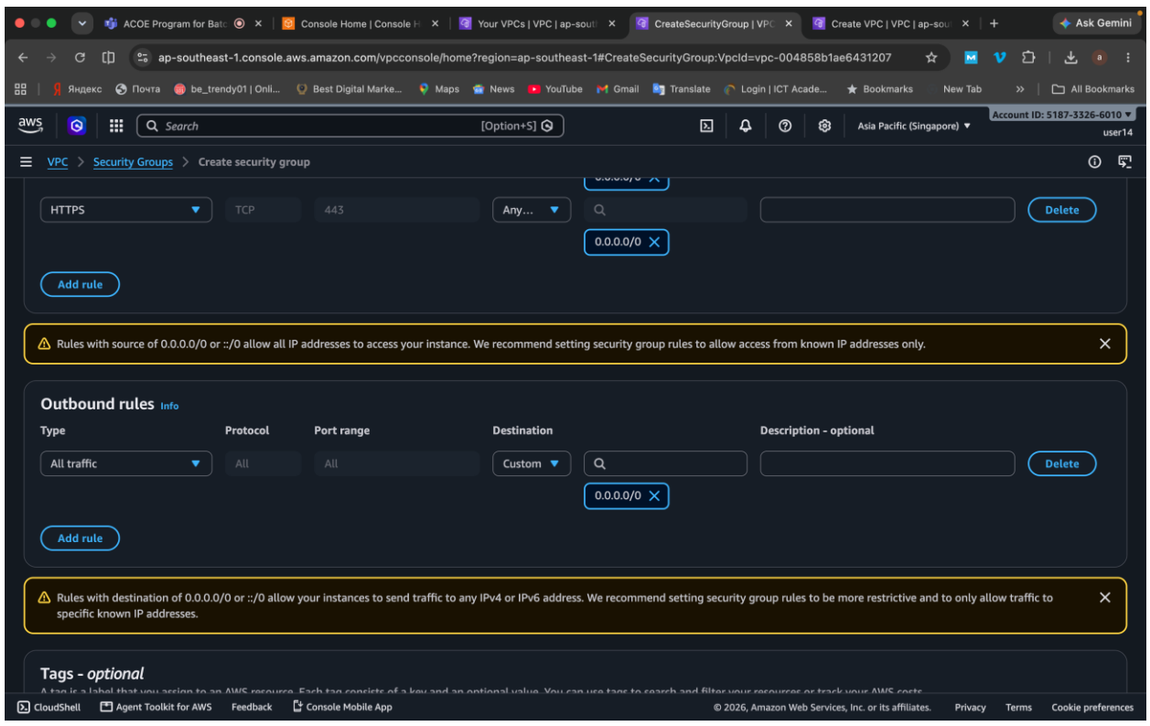

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user14
VPC > Your VPCs > Create VPC
Customize subnets CIDR blocks
NAT gateways ($) - updated Info
NAT gateway allows private resources to access the internet from any availability zone within a VPC, providing a single managed internet exit point for the entire region. Additional charges apply.
None Regional - new Zonal
Introducing regional NAT gateway
AWS now offers a multi-AZ NAT Gateway, eliminating the need for separate NAT Gateways across availability zones.
VPC endpoints Info
Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.
None S3 Gateway
DNS options Info
Enable DNS hostnames
Enable DNS resolution
Preview
VPC Show details
Your AWS virtual network
project-vpc
Subnets (4)
Subnets within this VPC
ap-southeast-1a
project-subnet-public1-ap-
project-subnet-private1-ap-
ap-southeast-1b
project-subnet-public2-ap-
project-subnet-private2-ap-
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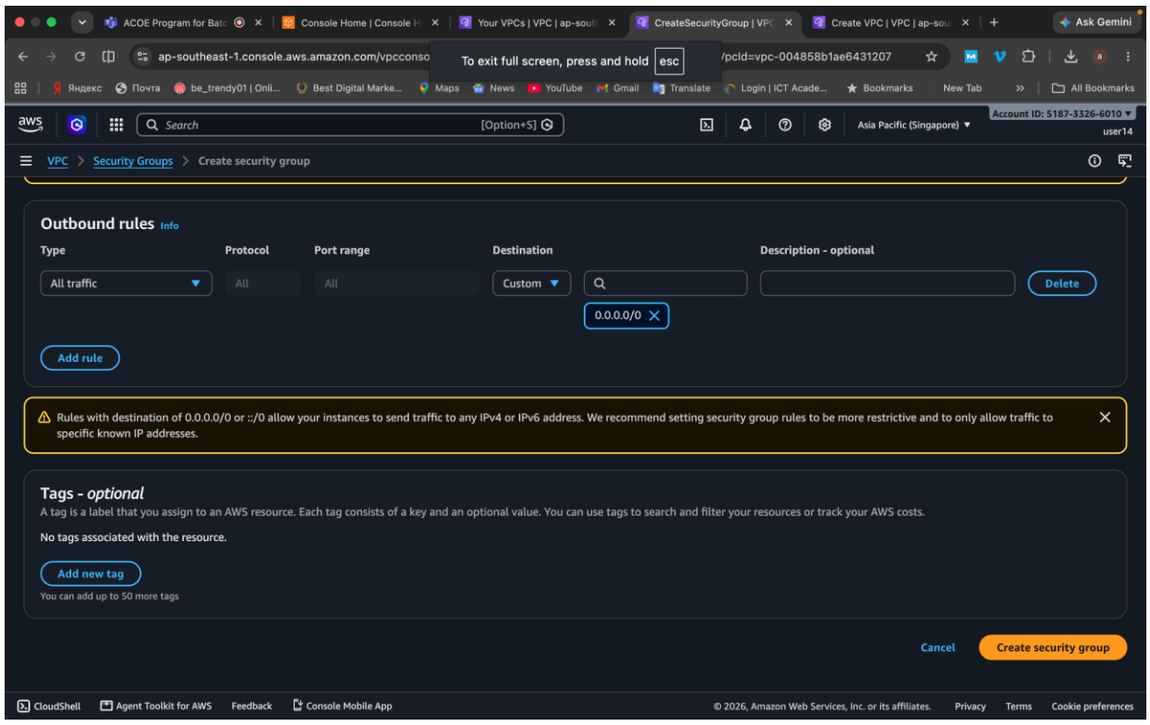


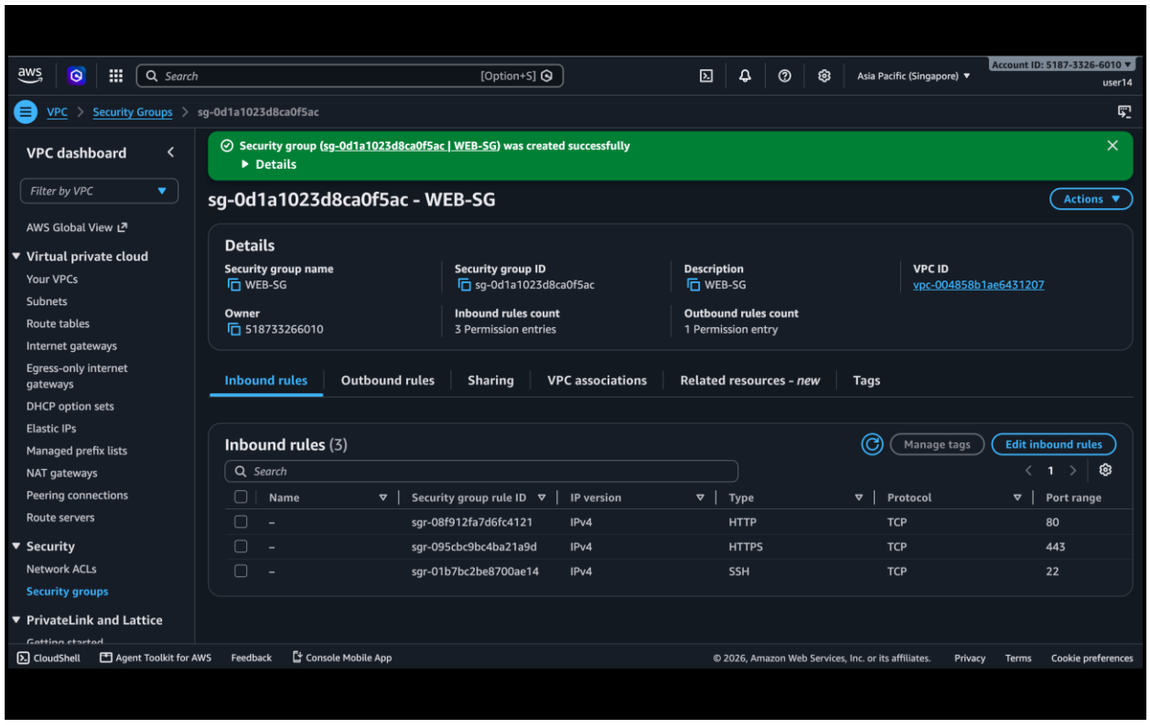
Task 2: Create 2 Security Groups

Creating a security group and giving name and description as WEB-SG and add rules as SSH, HTTP.

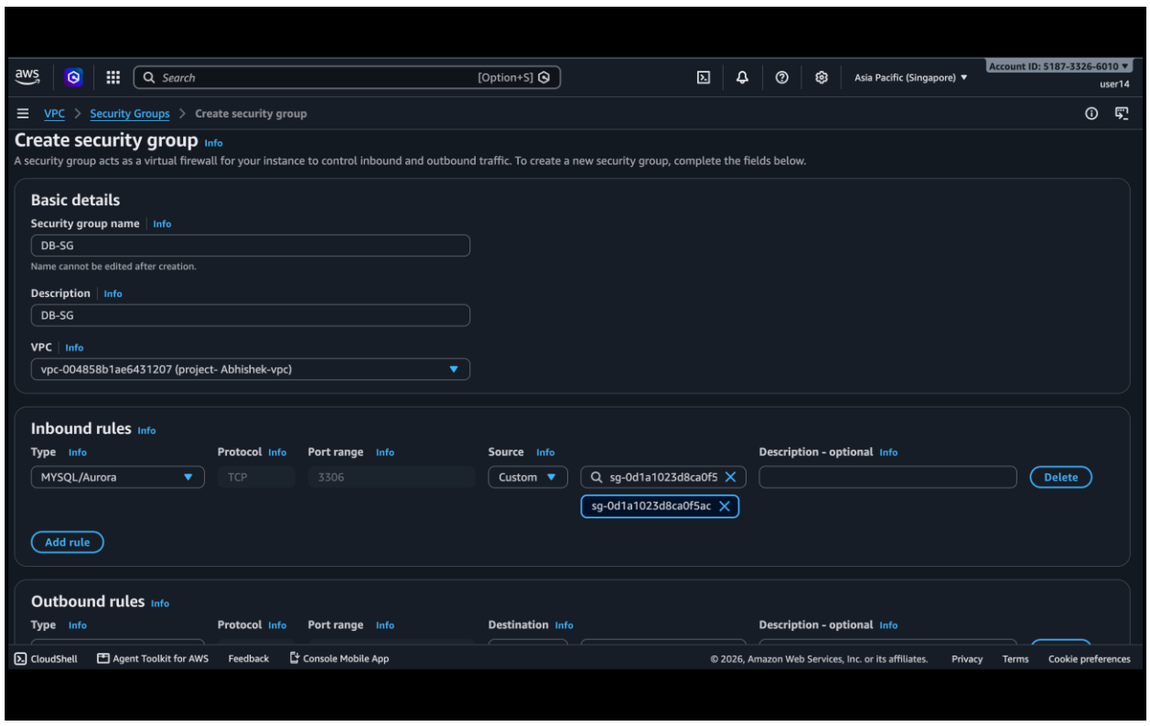


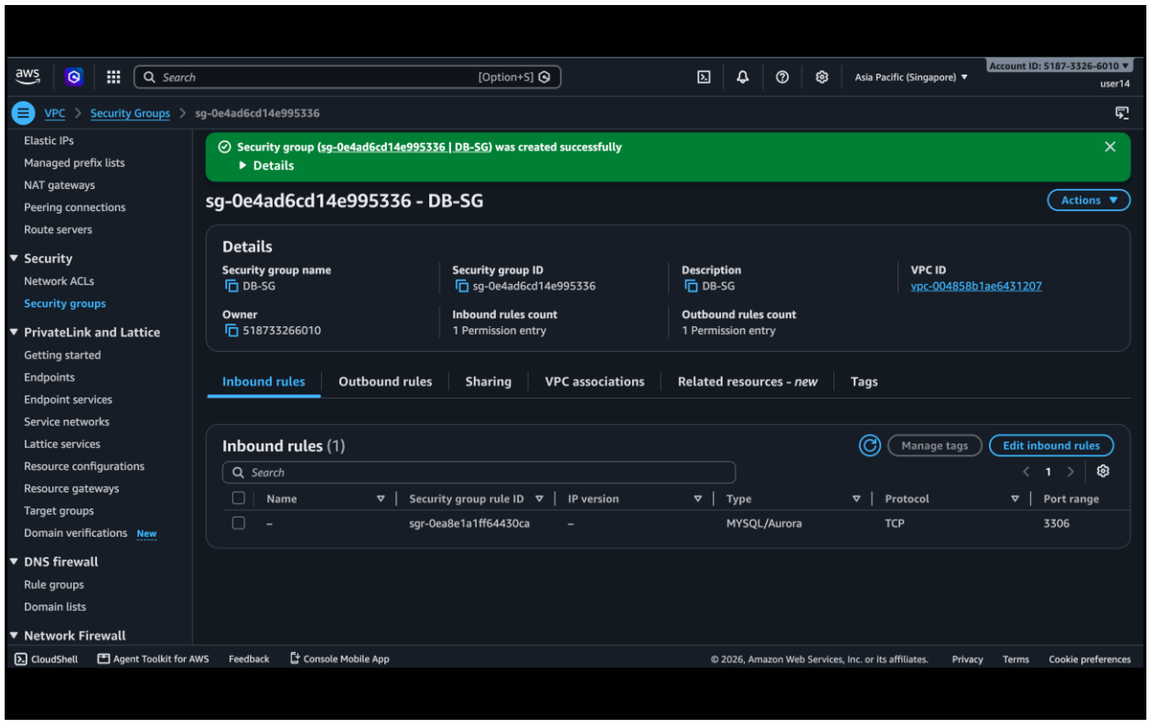

ap-southeast-1.console.aws.amazon.com/vpcconsole/home?region=ap-southeast-1#CreateSecurityGroup:VpcId=vpc-004858b1ae6431207
Search
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Asia Pacific (Singapore)
Account ID: 5187-3326-6010
user14
VPC > Security Groups > Create security group
HTTPS
TCP
443
Any...
0.0.0.0/0
Delete
Add rule
Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Outbound rules Info
Type
Protocol
Port range
Destination
Description - optional
All traffic
All
All
Custom
0.0.0.0/0
Delete
Add rule
Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.
Tags - optional
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Create one more security group for database. Go to VPC service and select security group from left hand side then click on create security group. Give name and description as DB-SG and select demo-vpc created in step 1. In the inbound rule section, click on **Add rule.**

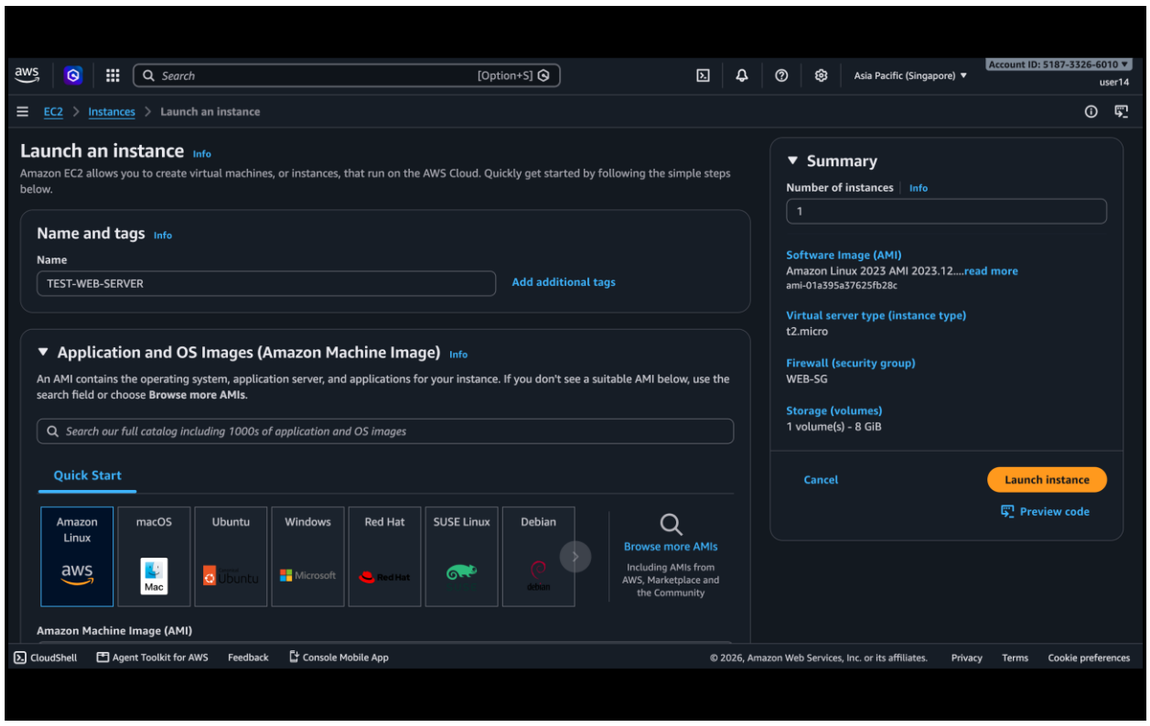




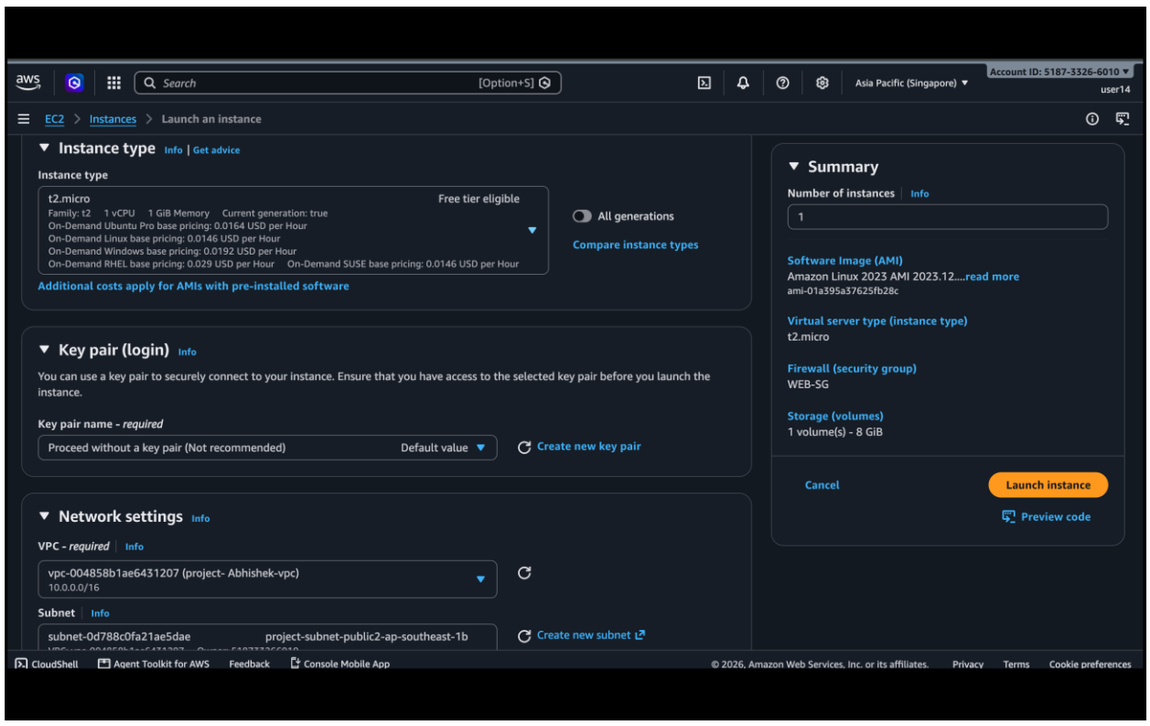
In Type select **MYSQL/Aurora**, select **Custom** and click on **WEB-SG** (security group created)

Task 3: Create, connect, and configure Instance

To launch 1 testing Instance which we will use to install our require packages to host the application, also we will test whether application running on this instance is connecting to RDS databases successfully and finally we will create our AMI image from it.
Going to the console and creating an EC2 with instance name TEST-WEB-SERVER (using Ubuntu AMI)

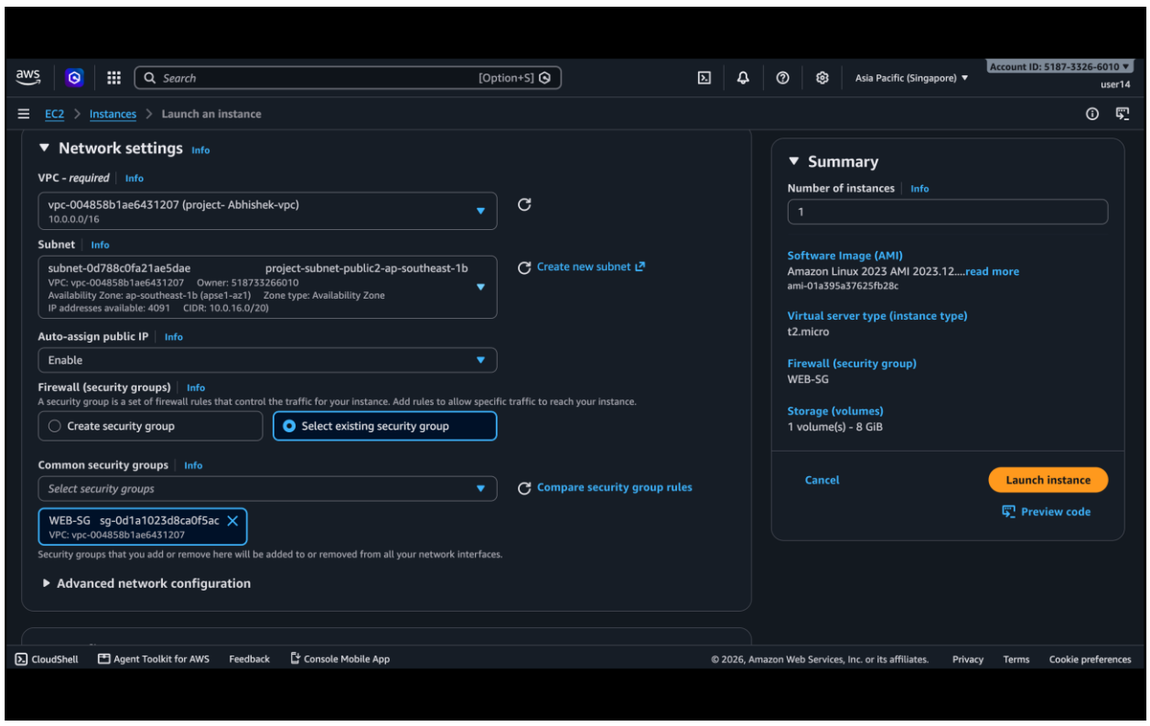


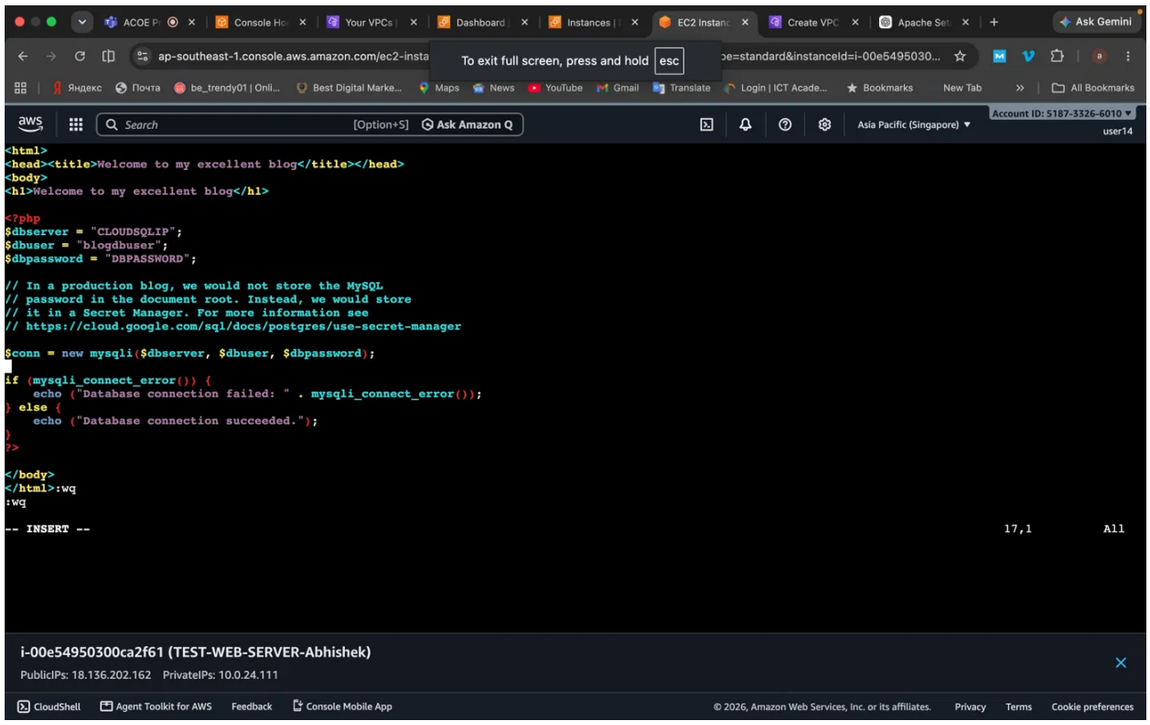
Under instance type – change it to Amazon Linux 2 AMI (HVM)
Under Key pair (login) – Select Proceed without a key pair



When you click on edit, the page below will be displayed. We are supposed to select the VPC which we have created with the name **Demo-vpc.**

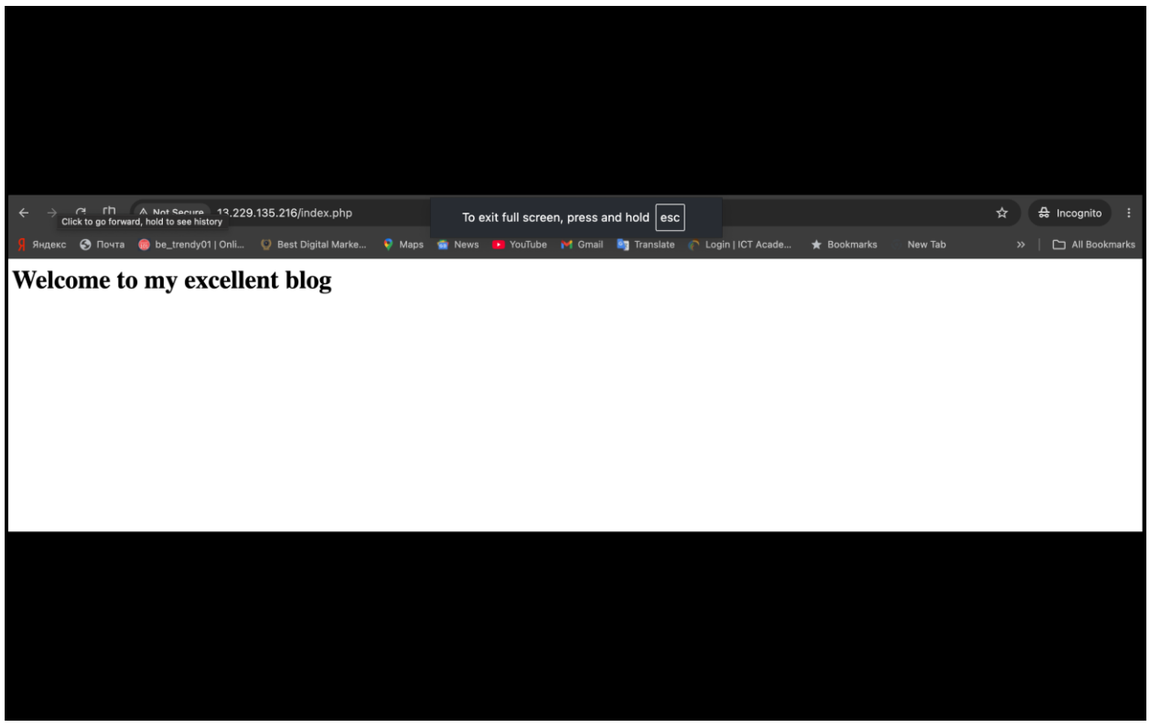
a) Under subnets we must select any one **public subnet** that we have created previously.
b) Under Auto assign public IP select enable.
c) Under Firewall (Security Groups) – Select **existing security group.**
d) Under Common Security Group- Select **WEB-SG**.





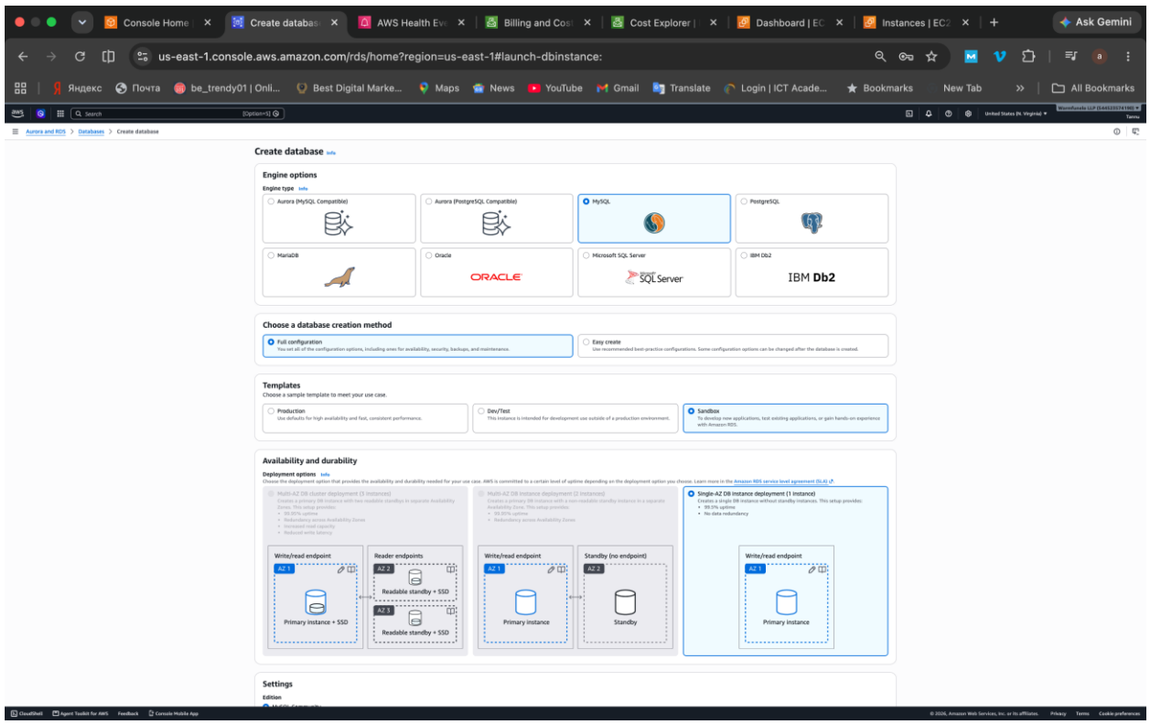
copy and paste the public IP of your instance in browser and add /index.php after the public IP of your instance. Below is the expected output. (Note that it is not displaying the

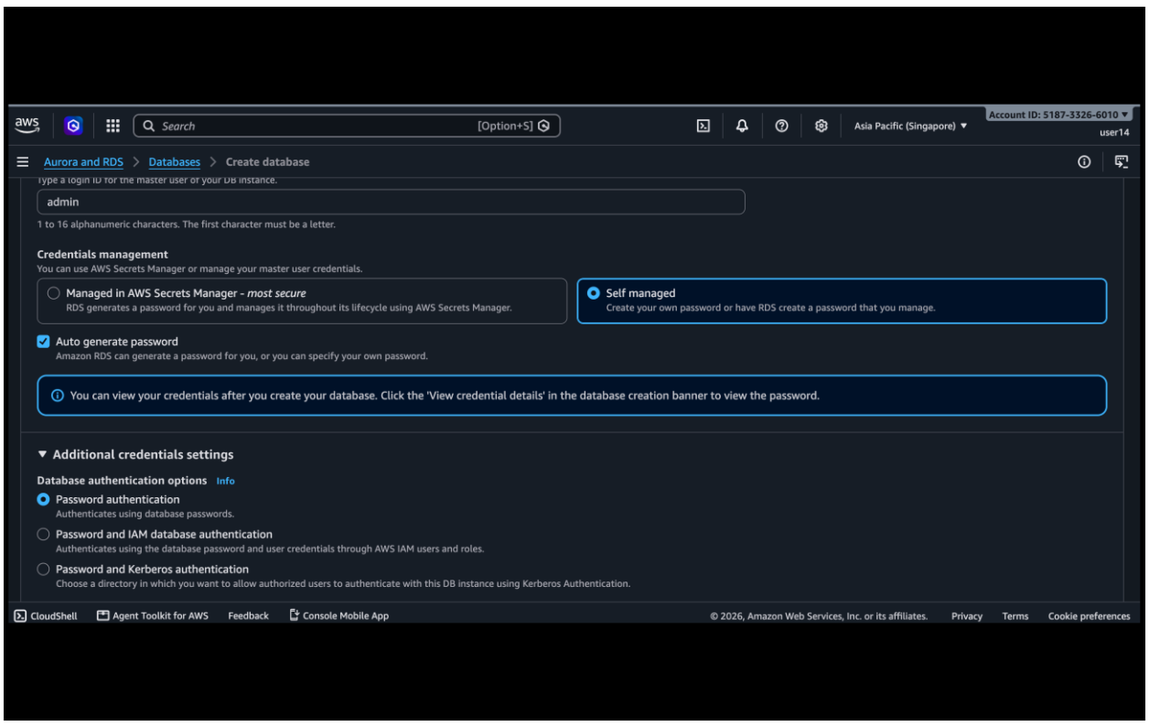
message that the database is connected successfully).

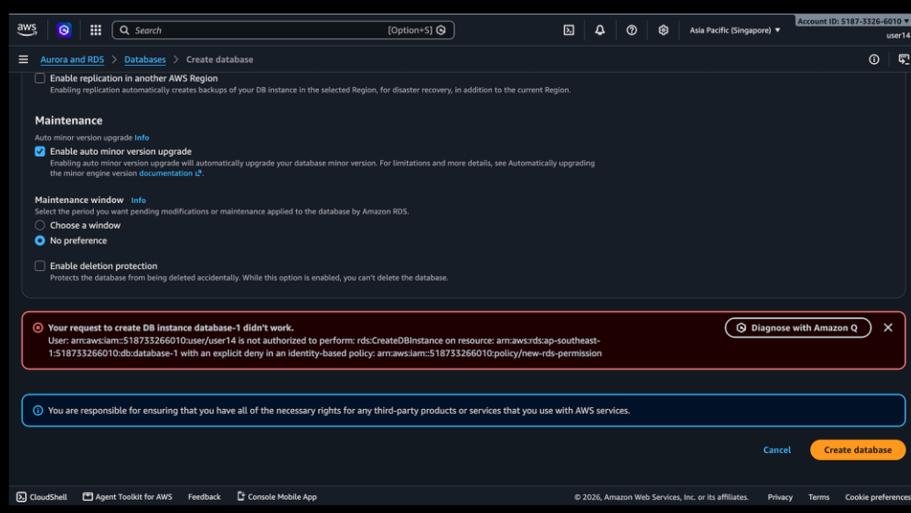


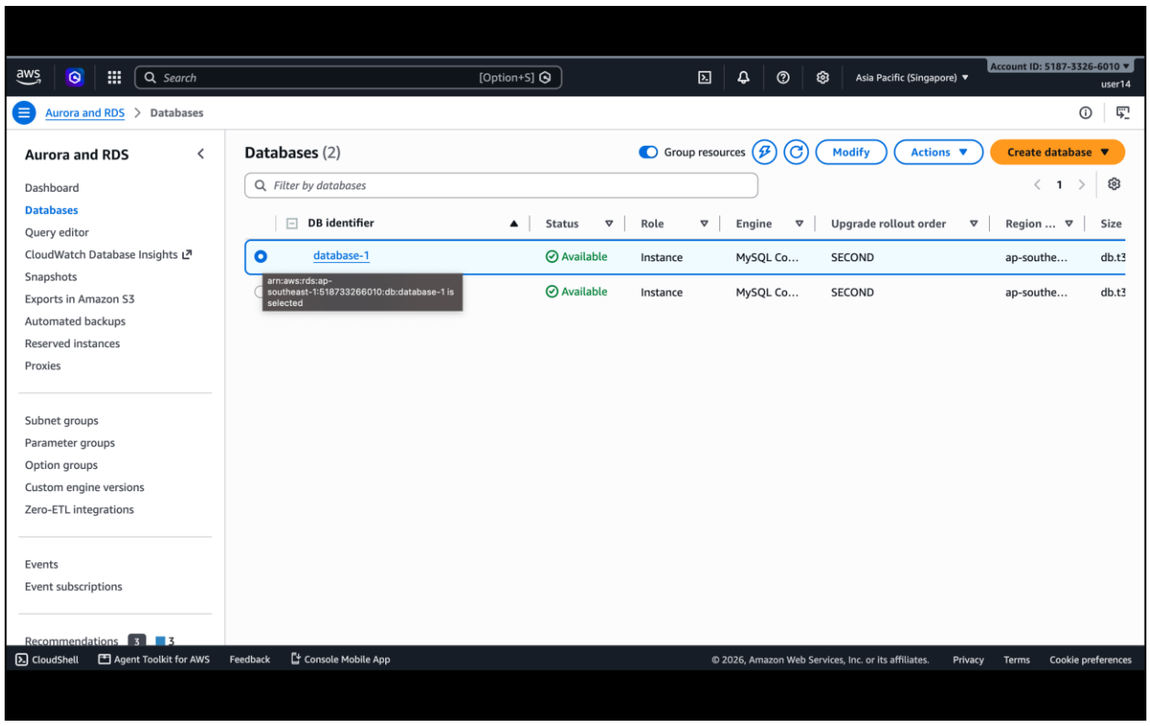
Task 4: Create, RDS MySQL database.

Select **Database** then click on **Create Database, use sandbox template and mysql DB**

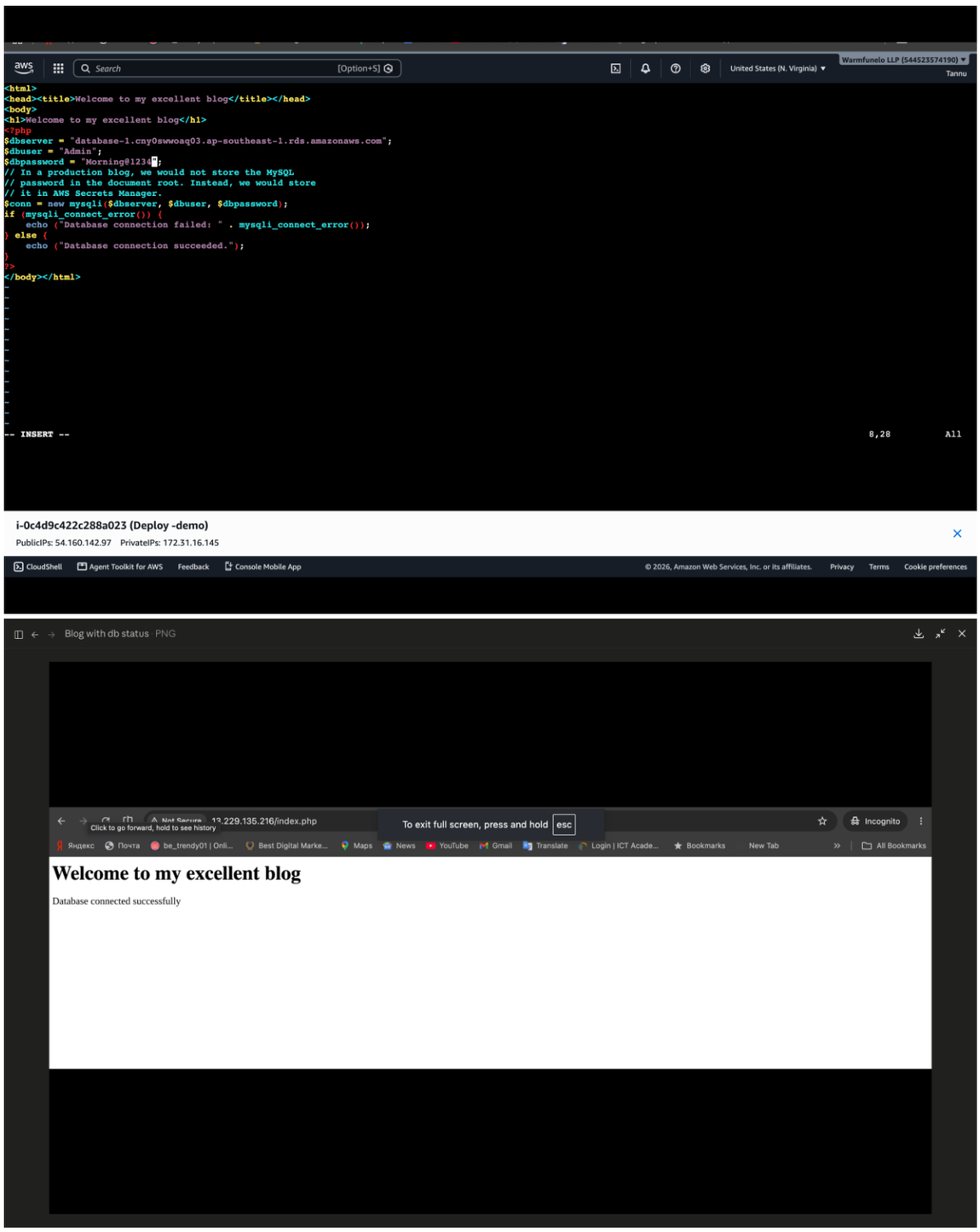



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Aurora and RDS
Databases
Create database
admin
1 to 16 alphanumeric characters. The first character must be a letter.
Credentials management
You can use AWS Secrets Manager or manage your master user credentials.
Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
Self managed
Create your own password or have RDS create a password that you manage.
Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.
You can view your credentials after you create your database. Click the 'View credential details' in the database creation banner to view the password.
Additional credentials settings
Database authentication options Info
Password authentication
Authenticates using database passwords.
Password and IAM database authentication
Authenticates using the database password and user credentials through AWS IAM users and roles.
Password and Kerberos authentication
Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.
CloudShell
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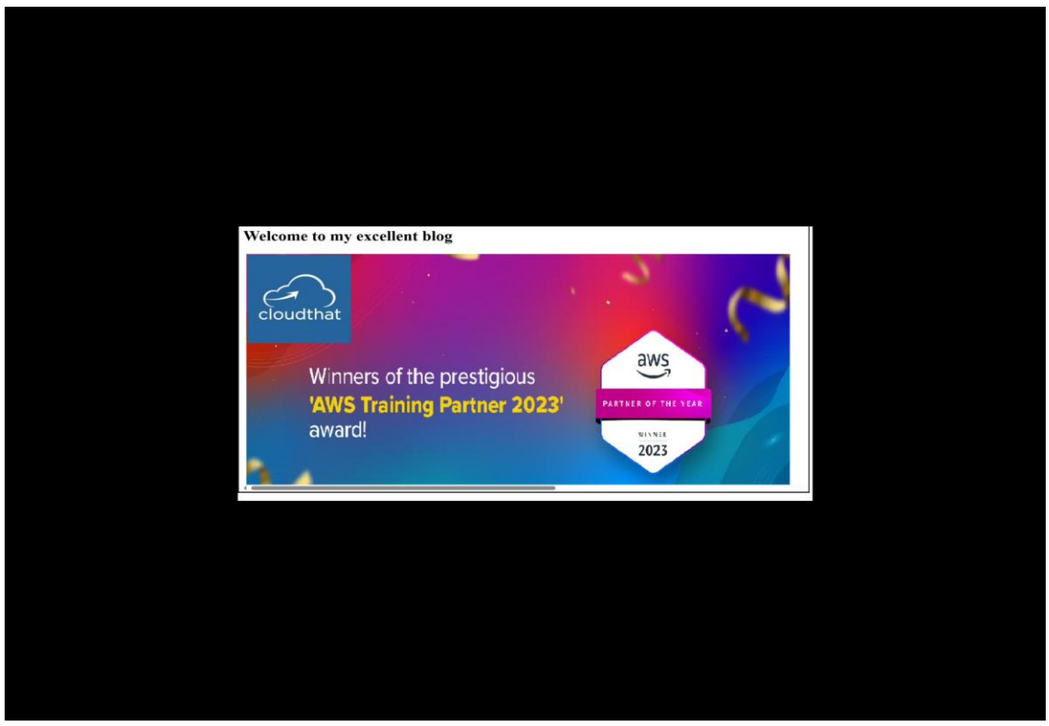

Search
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Asia Pacific (Singapore)
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user14
Aurora and RDS
Databases
Create database
Enable replication in another AWS Region
Enabling replication automatically creates backups of your DB instance in the selected Region, for disaster recovery, in addition to the current Region.
Maintenance
Auto minor version upgrade Info
Enable auto minor version upgrade
Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see Automatically upgrading the minor engine version documentation.
Maintenance window Info
Select the period you want pending modifications or maintenance applied to the database by Amazon RDS.
Choose a window
No preference
Enable deletion protection
Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.
Your request to create DB instance database-1 didn't work.
User: arn:aws:iam::518733266010:user/user14 is not authorized to perform: rds:CreateDBInstance on resource: arn:aws:rds:ap-southeast-1:518733266010:db:database-1 with an explicit deny in an identity-based policy: arn:aws:iam::518733266010:policy/new-rds-permission
Diagnose with Amazon Q
You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.
Cancel
Create database
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Task 5: Configure Instance to connect to MySQL Database



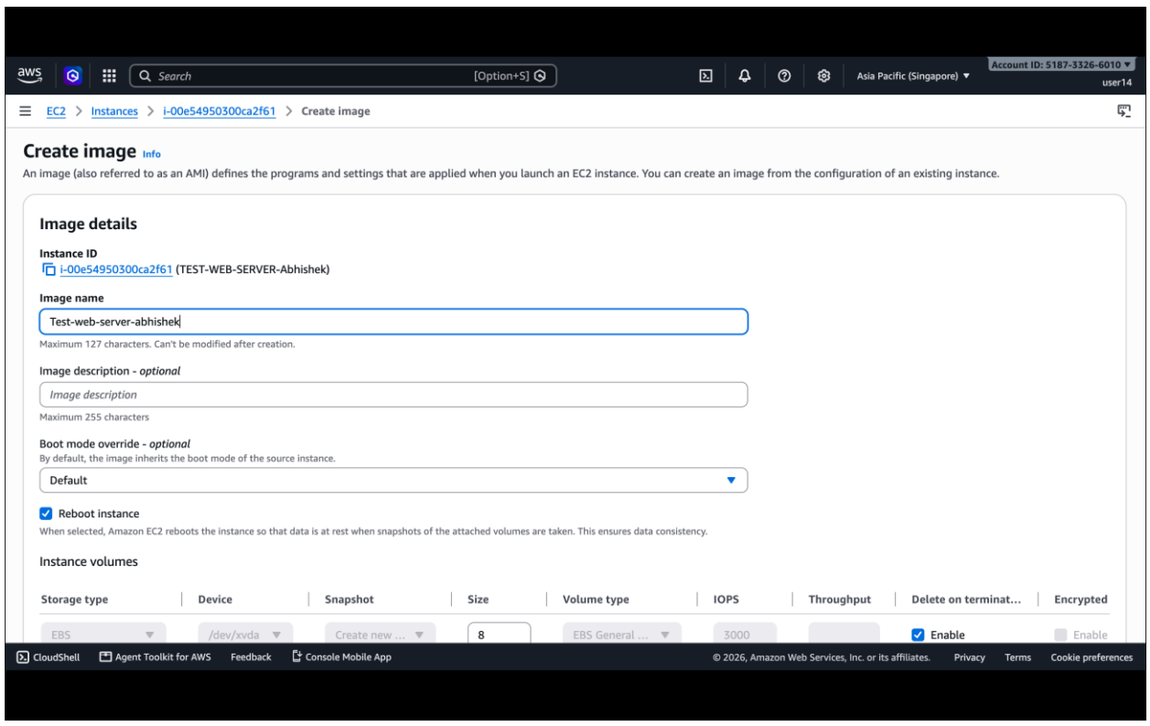
Task 6: Configure Instance to get the images from Amazon S3 on website

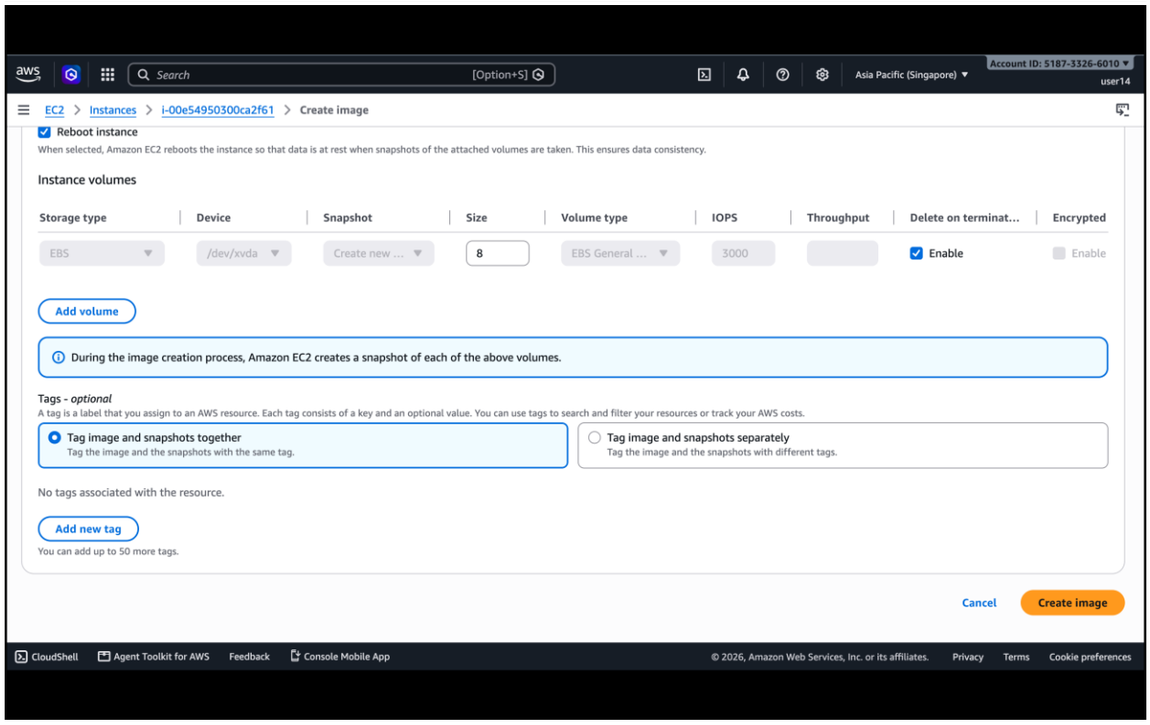


Task 7: Create AMI image of Instance.

In creating an Image window under **an Image** name, keep other option default and select **Create image.**

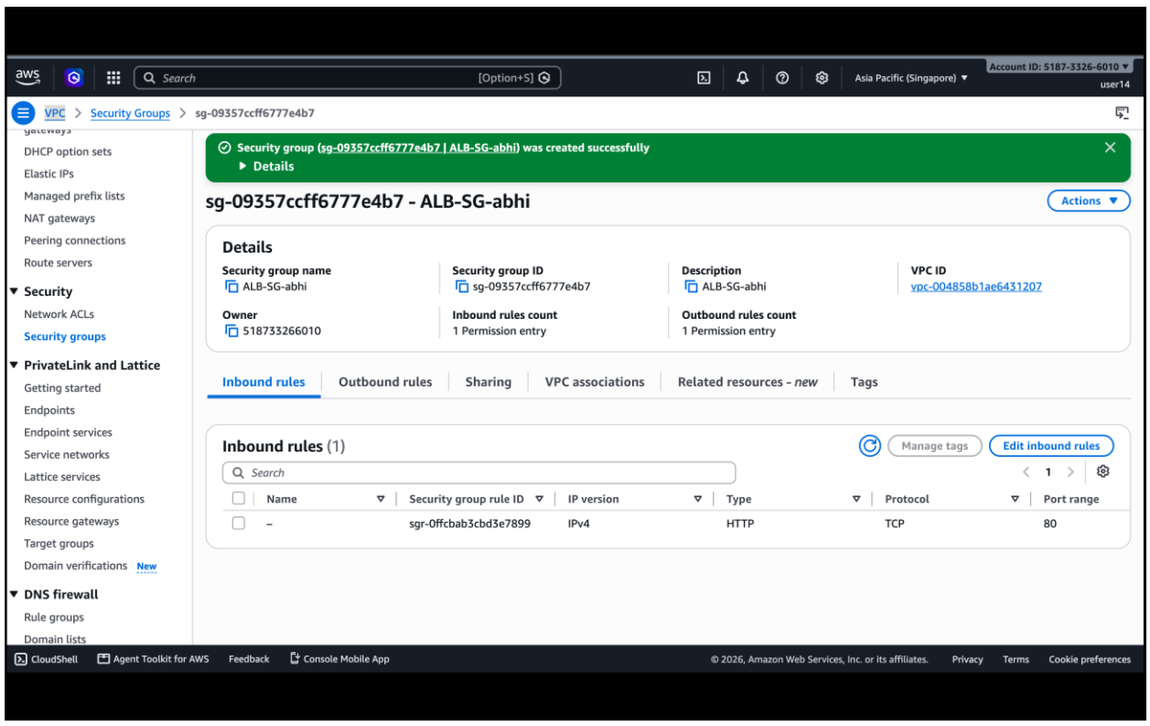
we will take AMI image of our server. For this go to EC2 service on console, select your instance, click on **Action**, and click on **Images and templates**. Click on **Create Image.**

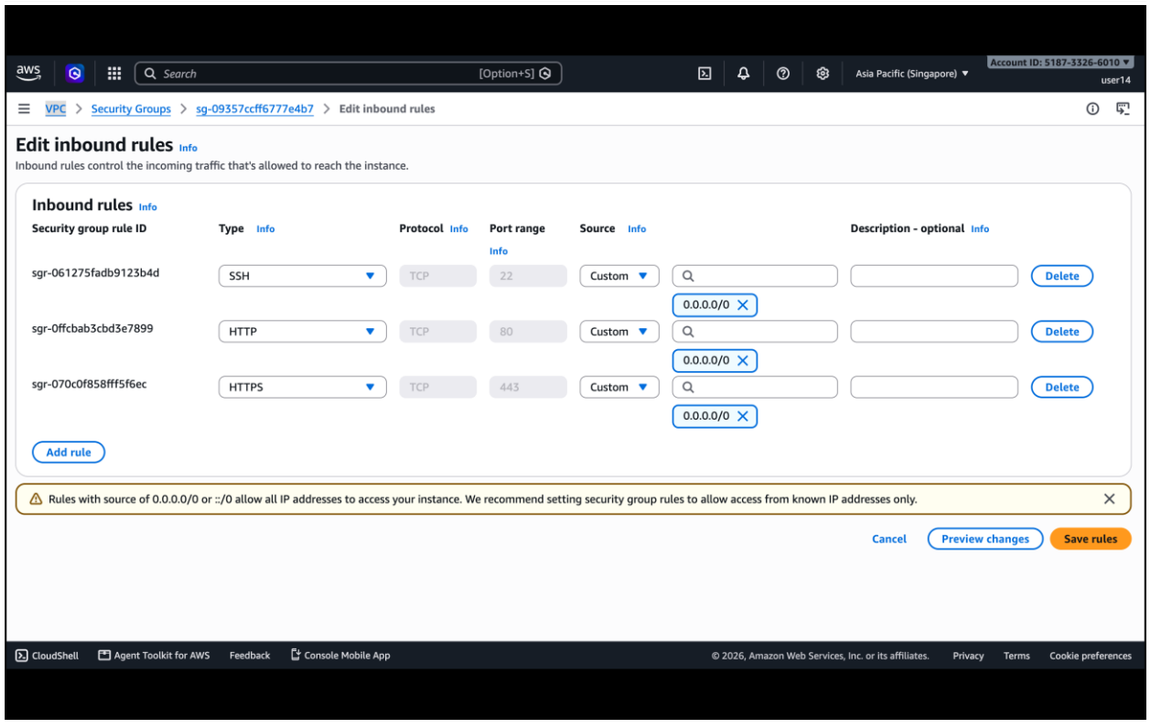




Task 8: Create security group for ALB and modify security group of instance.

Go to the AWS Management console and search for VPC and in the left-hand side we will find the Security Group. Give name and description as ALB-SG

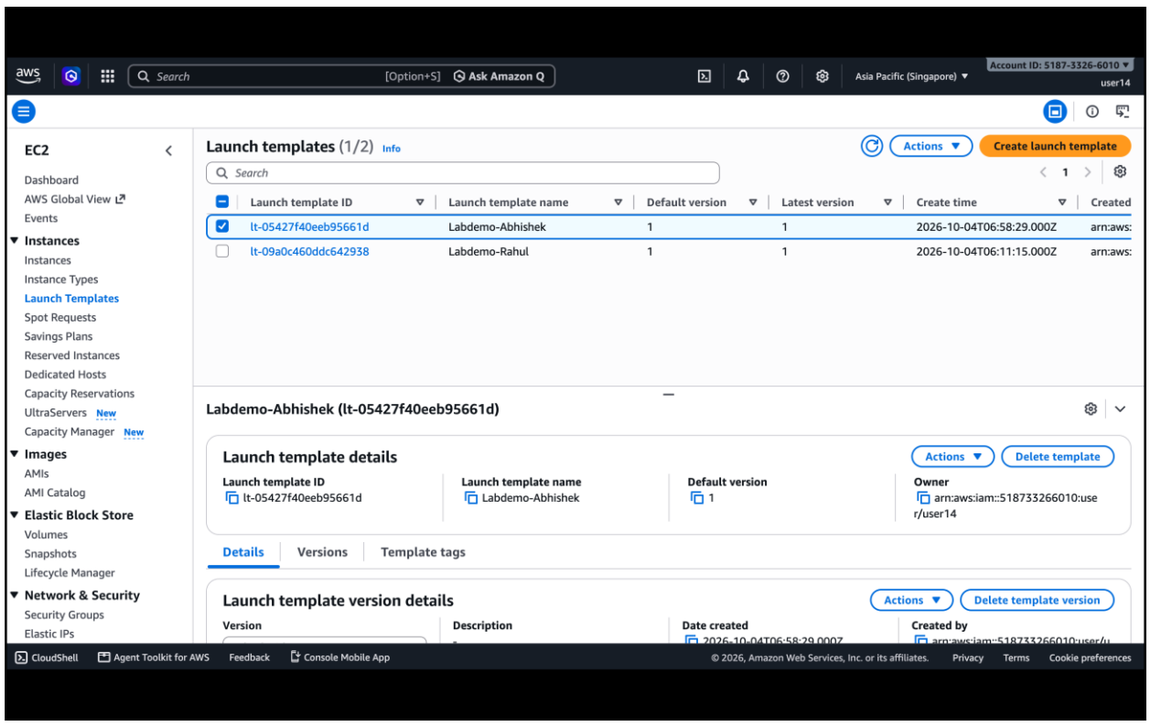




Task 9: Implement high availability and scalability to the application.

In create launch template window select below options:

- **Name**- Labdemo
- Under **Application and OS images** section select **My AMI**. Select the AMI created in task 7.
- **Instance Type**- t2.micro
- **Security Group**- WEB-SG
- **Key pair name-** Don't include in launch template.
- **Subnet-** Don't include in launch template.
- **Firewall- Select existing security group** then select WEB-SG
- Keep other option default and select **create launch template.**


Search
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EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Launch templates (1/2) Info
Search
Actions
Create launch template
Launch template ID
Launch template name
Default version
Latest version
Create time
Created
lt-05427f40eeb95661d
Labdemo-Abhishek
1
1
2026-10-04T06:58:29.000Z
arn:aws:
lt-09a0c460ddc642938
Labdemo-Rahul
1
1
2026-10-04T06:11:15.000Z
arn:aws:
Labdemo-Abhishek (lt-05427f40eeb95661d)
Launch template details
Actions
Delete template
Launch template ID
lt-05427f40eeb95661d
Launch template name
Labdemo-Abhishek
Default version
1
Owner
arn:aws:iam::518733266010:user/user14
Details
Versions
Template tags
Launch template version details
Actions
Delete template version
Version
Description
Date created
Created by
CloudShell
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